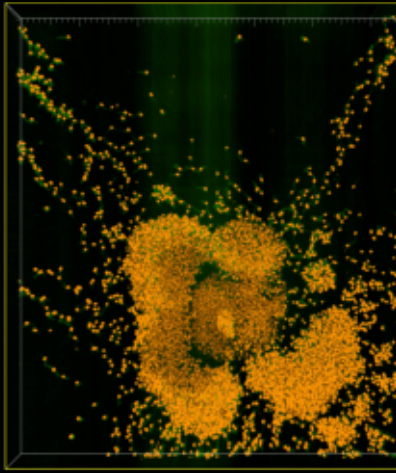
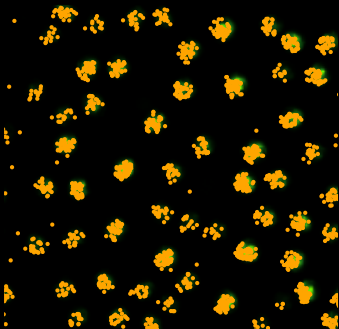


How the agentic loop discovered (and falsified) a Dictyostelium model
10 steps: agent DID -> FOUND -> final simulation. Bottom line: the model makes multi-mound only transiently and collapses; no 2D mechanism (12 tested) reproduces the real dense multi-mound aggregate.

REAL Dictyostelium data (target)
dense, multi-mound aggregate (~8 mounds)

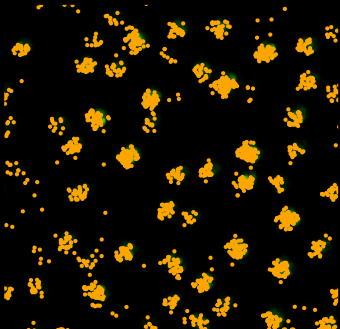


Influx + linear chemotaxis (secrete->sense), repulsion only



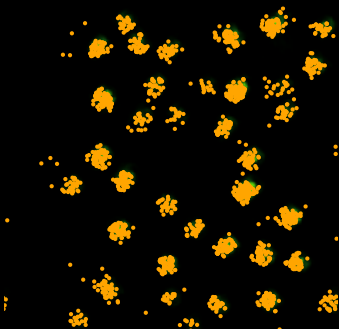
-> Cells drift; weak diffuse clump, no structure
[mounds=40, inner=0.21, N=1984]

Crank chemotaxis (gain high, low cAMP diffusion)



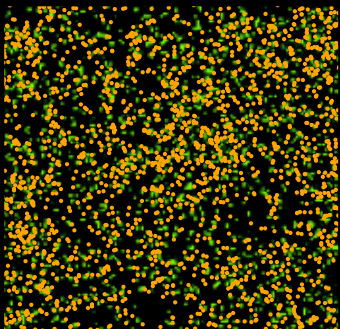
-> Keller-Segel collapse -> ONE central blob
[mounds=34, inner=0.20, N=1984]

Add spring adhesion (kadh high, long r_on)



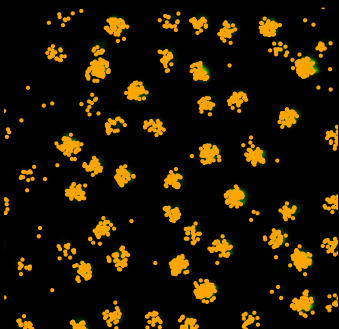
-> Filament / strand network, not compact mounds
[mounds=18, inner=0.19, N=1984]

Add excitable cAMP relay waves (relay op)



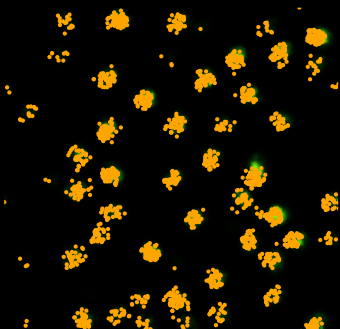
-> Still 1-few mounds; relay NOT sufficient
[mounds=55, inner=0.21, N=1984]

Replace sense with Hill-saturated chemotaxis (sense_sat)



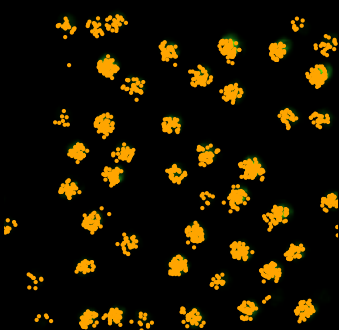
-> BREAKTHROUGH - first MULTI-MOUND (5-6 compact mounds)
[mounds=26, inner=0.19, N=2184]

Tune the sense_sat parent (D=0.0042, secrete=11)



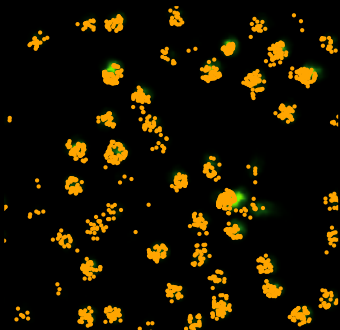
-> REAL-like 4-8 dense mounds across seeds (best match)
[mounds=20, inner=0.22, N=2200]

Fix the metric (SSIM-on-image + mound-count; inner_mass was gameable)



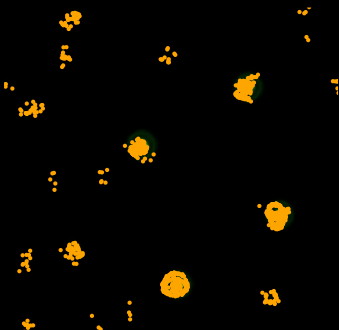
-> Revealed real target ~8 mounds; old metric hid morphology
[mounds=30, inner=0.19, N=2200]

Add density_repel (finite cell volume)



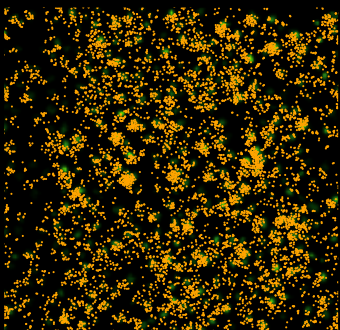
-> Rescues high-density runaway but no stable mounds
[mounds=3, inner=0.22, N=2200]

Stress test: run far longer (n_frames 400 -> 2000)



-> HEADLINE: multi-mound is a TRANSIENT -> collapses to a single blob
[mounds=4, inner=0.00, N=2600]

Fork to soft-MPM bodies (8 particles/cell)



-> Diffuse, inner~0.2 - deformable body NOT the missing ingredient
[mounds=54, inner=0.21, N=1148]